Lesson 2.5

Two-Column Proofs

Statements on the left, reasons on the right

What a two-column proof looks like

Statements	**Reasons**
1. . . . (given)	Given
2. . . .	a property, definition, . . .
. . . (what you must prove)	. . .

What a two-column proof looks like

Statements	**Reasons**
1. . . . (given)	Given
2. . . .	a property, definition, . . .
. . . (what you must prove)	. . .

Start with what is **given**. End with what you must **prove**. Every line must follow from the lines above it.

An algebra proof

Given: $2x + 5 = 13$ Prove: $x = 4$

An algebra proof

Given: $2x + 5 = 13$ Prove: $x = 4$

1. $2x + 5 = 13$	Given

An algebra proof

Given: $2x + 5 = 13$ Prove: $x = 4$

1. $2x + 5 = 13$	Given
2. $2x = 8$	Subtraction Property of Equality

An algebra proof

Given: $2x + 5 = 13$ Prove: $x = 4$

1. $2x + 5 = 13$	Given
2. $2x = 8$	Subtraction Property of Equality
3. $x = 4$	Division Property of Equality

A geometry proof

Given: $AB = CD$ Prove: $AC = BD$



A geometry proof

Given: $AB = CD$ Prove: $AC = BD$

1. $AB = CD$	Given
2. $BC = BC$	Reflexive Property



A geometry proof



Given: $AB = CD$ Prove: $AC = BD$

1. $AB = CD$	Given
2. $BC = BC$	Reflexive Property
3. $AB + BC = BC + CD$	Addition Property of Equality

A geometry proof



Given: $AB = CD$ Prove: $AC = BD$

1. $AB = CD$ Given
2. $BC = BC$ Reflexive Property
3. $AB + BC = BC + CD$ Addition Property of Equality
4. $AB + BC = AC$, $BC + CD = BD$ Segment Addition Postulate

A geometry proof



Given: $AB = CD$ Prove: $AC = BD$

1. $AB = CD$ Given
2. $BC = BC$ Reflexive Property
3. $AB + BC = BC + CD$ Addition Property of Equality
4. $AB + BC = AC$, $BC + CD = BD$ Segment Addition Postulate
5. $AC = BD$ Substitution Property

Watch out

Watch out! A reason must be a **given**, a **definition**, a **property**, a **postulate** or a **theorem** — never "it looks like it" or "I calculated it".

Your turn! Pause the video and try it.

Given: $5x - 2 = 18$ Prove: $x = 4$

1. $5x - 2 = 18$ Given 2. $5x = 20$? 3. $x = 4$?

Your turn! Pause the video and try it.

Given: $5x - 2 = 18$ Prove: $x = 4$

1. $5x - 2 = 18$ Given 2. $5x = 20$? 3. $x = 4$?

2. Addition Property of Equality 3. Division Property of Equality

Remember

- **Statements** on the left, **reasons** on the right.
- Start with the **given**; end with what you must **prove**.
- Every reason is a given, definition, property, postulate or theorem.

Now practice at your level:

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